

6	(a) (i)	energy = EQ $= 9.0 \times 22 \times 10^{-3}$ $= 0.20 \text{ J}$	C1	A1 [2]
	(ii)	1. $C = Q/V$ $V = (22 \times 10^{-3})/(4700 \times 10^{-6})$ $= 4.7 \text{ V}$	C1	A1 [2]
	2.	<i>either</i> $E = \frac{1}{2}CV^2$ $= \frac{1}{2} \times 4700 \times 10^{-6} \times 4.7^2$ $= 5.1 \times 10^{-2} \text{ J}$	C1	A1 [2]
	<i>or</i>	$E = \frac{1}{2}QV$ $= \frac{1}{2} \times 22 \times 10^{-3} \times 4.7$ $= 5.1 \times 10^{-2} \text{ J}$	(C1)	(A1)
	<i>or</i>	$E = \frac{1}{2}Q^2/C$ $= \frac{1}{2} \times (22 \times 10^{-3})^2/4700 \times 10^{-6}$ $= 5.1 \times 10^{-2} \text{ J}$	(C1)	(A1)

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- (b)** energy lost (as thermal energy) in resistance/wires/battery/resistor B1 [1]
(award only if answer in (a)(i) > answer in (a)(ii) 2)